

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – CONCEPT MAPPING

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 2 – Concept Mapping
Weightage:	20% of CIA	Total Marks:	100 (2 Questions × 50 Marks)
CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:	Naveen Kumar S	Reg. No.:	192421030
Section / Batch:	B Tech IT	Date:	24th July 2026

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Draw a clear and neat concept map for each question.
- Identify all key concepts and arrange them in a logical hierarchy.
- Show meaningful linkages between concepts using arrows and connecting words.
- Compare related concepts wherever required and justify the relationships clearly.
- Ensure the concept map is well-organized, readable, and properly labelled.

Question Type	Focus Area	Questions	Marks
Concept Mapping	Map algorithm efficiency concepts using asymptotic analysis, search techniques, recurrence relations, and recursion tree method	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – CONCEPT MAPPING

Q25. Consider the concept map: Recurrence Relation → Expansion → Pattern Identification → Summation → Final Complexity. Explain how expansion helps uncover the repeated structure hidden inside a recurrence relation. Extend the map by adding Substitution Method as a linked analytical technique and show how it uses the identified pattern. Justify why recognizing the summation pattern is important in deriving the final complexity. Interpret how this map supports the systematic solving of recurrences without directly applying the Master Theorem.

Q26. Consider the concept map: Divide and Conquer → Problem Size Reduction → Number of Subproblems → Recurrence Relation → Complexity Explain how dividing the original problem into smaller subproblems leads naturally to a recurrence relation. Extend the map by adding the parameters a and b as children of Number of Subproblems and Problem Size Reduction respectively. Justify how these parameters influence the final complexity. Interpret how the concept map connects algorithm design structure with mathematical analysis.

Code of Conduct

I Naveen Kumar S (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Naveen Kumar S

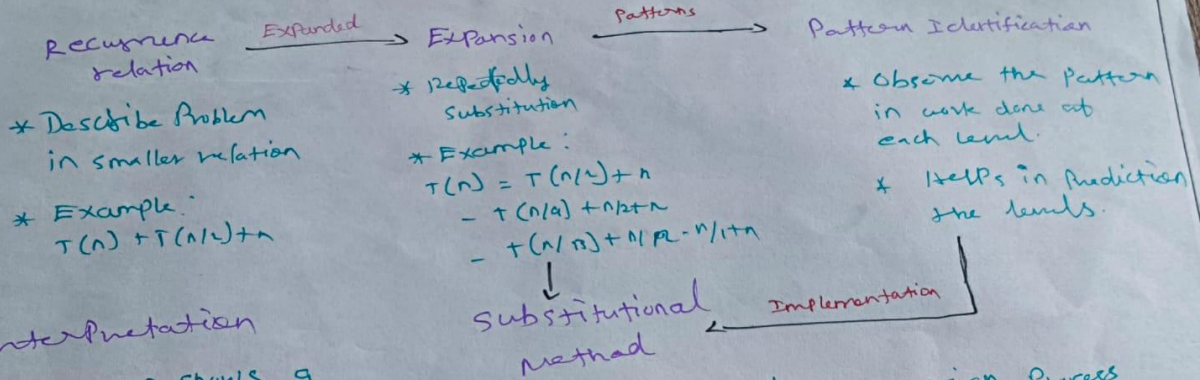
RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

Marks Evaluation:

Question No.	Concept Identification (25%)	Linkages (25%)	Organization (20%)	Integration of Literature (20%)	Clarity (10%)	Total Marks (Each – 50)
Q1						
Q2						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____

Q 25- Concept Map: Recurrence Relation $\rightarrow$ Expansion - Pattern Identification

Interpretation

This map shows a systematic way to solve recurrences by recognising the hidden repeated structure through expansion & pattern recognizing.

↳ use by Expansion Process

↳ Proves and supports functions

Recognizing Importance

- * it avoid evaluating work repeatedly
- * It gives a precise expression for total work.

Q2b. Concept Map: Divide and Conquer $\rightarrow$ Problem size reduction $\rightarrow$ No. of subProblems $\rightarrow$ Recurrence $\rightarrow$ ComplexityDivide & Conquer $\xrightarrow{\text{creates}}$ Problem size reduction $\xrightarrow{\text{decides}}$ No. of sub-Problems

- * Breaks the original Problem into smaller fragments

- * Solve each subProblems separately and combine the results

- * Each subProblem is of smaller size
- * size reduces by γ factor of b

- * At each step, the Problem is divided.
- * Number depends on Parameters

$\xrightarrow{\text{leads to}}$ Recurrence relation $\xrightarrow{\text{gives}}$ Complexity

- * Relation is $T(n)$ and $T(n/2)$

- * General form

$$\Rightarrow T(n) = a \cdot T(n/2) + T(n).$$

- * Solve the recurrence to find the order of growth
- * Represents the efficiency of the algorithm.